AQ9.5: Activity Questions 5 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Let $f(x, y, z) = (xy, x + y + z^2)$, where $x, y, z \in \mathbb{R}$. Choose the set of correct options.

- [ ] The domain of f is $\mathbb{R}^2$ and the co-domain of f is $\mathbb{R}^2$
- [ ] The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$
- [ ] f is continuous everywhere in its domain
- [ ] f is not continuous anywhere in its domain

No, the answer is incorrect.
Score: 0
Accepted Answers:

The domain of f is $\mathbb{R}^3$ and the co-domain of f is $\mathbb{R}^2$

f is continuous everywhere in its domain

1 point

Let $f(x, y) = \begin{cases} \frac{\cos y \sin x}{x} & x \neq 0 \\ \cos y & x = 0 \end{cases}$.

Choose the correct options.

- [ ] f is not defined at $(0, 0)$
- [ ] f is defined at $(0, 0)$
- [ ] f is continuous at $(0, 0)$
- [ ] f is not continuous at $(0, 0)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

f is defined at $(0, 0)$

f is continuous at $(0, 0)$

1 point

Let $h(u, v) = \sin(uv)$, $g(x) = x^3$, $f(y) = \cos y$. Let $p(x, y) = h(g(x), f(y))$

Choose the correct option(s).

- [ ] p is a scalar-valued function
- [ ] p is a vector-valued function
- [ ]

p is continuous everywhere since $g.f$ is continuous everywhere in its domain

☐

p is continuous everywhere, g is also continuous everywhere while f is not continuous everywhere in its domain

No, the answer is incorrect.
Score: 0
Accepted Answers:

p is a scalar-valued function

p is continuous everywhere since $g.f$ is continuous everywhere in its domain

1 point

Let $p(t) = (ln(1-t), \frac{1}{t}, 3t)$.

Choose the incorrect option(s).

☐

p is continuous for all real t less than 1 but not equal to zero

☐

p is continuous on R

☐

p is continuous in $\{t \in R | t < 1\}$

☐

p is continuous in $\{t \in R | t \neq 1, t \neq 0\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

p is continuous on R

p is continuous in $\{t \in R | t < 1\}$

p is continuous in $\{t \in R | t \neq 1, t \neq 0\}$

1 point

Let $f(x, y) = ln(x^2 + y^2 - 1)$. Identify where f is continuous. Note: Equation of a circle with radius r and center (a, b) is given by $(x-a)^2 + (y-b)^2 = r^2$

○ Everywhere in R^2

○ Everywhere in $R^2 \setminus \{(1, 0), (0, 1)\}$

○ Everywhere in R^2 outside the circle with radius 1 and center $(0, 0)$

○ None of the above

No, the answer is incorrect.
Score: 0
Accepted Answers:

Everywhere in R^2 outside the circle with radius 1 and center $(0, 0)$

Level 2

1 point

Let $g(u, v, w) = \frac{uv}{w}$ and $f(x, y, z) = g(P(x, y, z), Q(x, y, z), R(x, y, z))$

where $P(x, y, z) = e^{x^2 + y}$, $Q(x, y, z) = \sqrt{y^2 + z^2 + 3}$, $R(x, y, z) = \sin(xyz) + 5$

P(x, y, z) = ex2+y, Q(x, y, z) = $\sqrt{}$ y2 + z2 + 3

, R(x, y, z) = sin(xyz) + 5.

Choose the correct option(s).

☐ f f is a vector-valued function

☐ f f is a scalar-valued function

☐ f f is continuous everywhere in its domain.

☐ P P is continuous everywhere in its domain, while Q Q is not continuous everywhere in its domain

No, the answer is incorrect.
Score: 0
Accepted Answers:

f f is a scalar-valued function

f f is continuous everywhere in its domain.

1 point

Let

$f: R^2 \rightarrow R, f(x, y) = \begin{cases} \frac{x^2}{x^2+y^2} & (x, y) \neq (0,0) \\ 0 & (x, y) = (0,0) \end{cases}$ f : R2 → R, f(x, y) = {x2+y2x20(x, y) = (0, 0)(x, y) = (0, 0).

Choose the correct option(s).

☐ $\lim_{(x,y)\rightarrow(0,0)} f$ lim(x,y)→(0,0) f does not exist

☐ $\lim_{(x,y)\rightarrow(0,0)} f$ lim(x,y)→(0,0) f exists

☐ f f is continuous at $(0,0)$ (0, 0)

☐ f f is not continuous at $(0,0)$ (0, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\lim_{(x,y)\rightarrow(0,0)} f$ lim(x,y)→(0,0) f does not exist

f f is not continuous at $(0,0)$ (0, 0)

1 point

Let

$f: R^2 \rightarrow R, f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & (x, y) \neq (0,0) \\ 0 & (x, y) = (0,0) \end{cases}$ f : R2 → R, f(x, y) = {x2+y2xy0(x, y) = (0, 0)(x, y) = (0, 0).

Choose the correct option(s).

☐ $\lim_{(x,y)\rightarrow(0,0)} f$ lim(x,y)→(0,0) f does not exist

☐ $\lim_{(x,y)\to(0,0)} f$ exists

☐ f is continuous at $(0,0)$

☐ f is not continuous at $(0,0)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\lim_{(x,y)\to(0,0)} f$ does not exist
f is not continuous at $(0,0)$

1 point

Let $f(r) = \frac{1}{<r,r>-1}$, where $r = (x, y, z)$ and the inner product $<,>$ is the dot product
Choose the set of incorrect options.

☐ f is continuous throughout R^3

☐ f is continuous throughout $R^3 \setminus \{(0,0,0)\}$

☐ f is discontinuous on the unit sphere $\{r \in R^3 : \| r \| = 1\}$

☐ f is continuous throughout $R^3 \setminus \{(1,0,0)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
f is continuous throughout R^3
f is continuous throughout $R^3 \setminus \{(0,0,0)\}$
f is continuous throughout $R^3 \setminus \{(1,0,0)\}$

Let $f : R^2 \to R$, $f(x,y) = \begin{cases} \frac{x^k y}{x^4 + y^4} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$. Find the minimum possible positive integer value of k such that f is continuous at (0,0).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4

1 point

Check Answers

Your score is: 0/10